

Problem Statement - 5

Title of Problem Statement: AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Background: Bitcoin's pseudonymous, peer-to-peer design lets criminal actors move, layer, and cash out illicit funds — ransomware payments, darknet-market proceeds, extortion, and laundering — while evading traditional financial surveillance.

The objective of problem statement is to design and build a complete system (offline) that ingests bulk Bitcoin transaction/network metadata (in CSV/JSON/XML), correlates network-layer (IP/port/timing) observations with blockchain-layer (wallet/TXID/amount) data, and applies AI/ML to detect anomalies, cluster entities, and generate prioritized, explainable investigative leads.

Description:

i. Challenge Objectives-

- Ingest & parse a bulk metadata dataset (timestamp, src/dst IP & port, TXID, input/output wallet addresses, amounts, fee, script type).
- Build an entity/transaction graph linking IPs, wallets, and transactions.
- Implement AI/ML detection use case (see Section 4) with a working model — not just rules.
- Generate a ranked, explainable alert list (why a wallet/transaction was flagged, with a confidence score).
- Present findings via a simple dashboard or link-analysis visualization.

ii. Suggested AI/ML Focus Areas

Focus Area	What to Build
Entity Clustering	Group wallets likely owned by one entity using common-input-ownership + graph embeddings.
Anomaly Detection	Flag statistically unusual transactions/flows
Peeling-Chain/ Detection	Mixing Detect laundering-pattern transaction sequences (peeling chains, CoinJoin-like structures).
Risk Scoring	Propagate risk scores from seed illicit wallets via algorithms

iii. Dataset: Parameters & Synthetic Generation

Participants will work with a synthetic dataset modelled on real Bitcoin P2P/transaction fields (no real seized or live-intercept data will be provided). Minimum fields: timestamp, src_ip, dst_ip,

src_port, dst_port, txid, input_addresses[], output_addresses[], input_amounts[], output_amounts[], geo_country/asn (integrate open source downloadable Geo IP database).

Expected Solution/Deliverables:

- Workable complete offline solution for linux platform.
- Working prototype (code repo) with ingestion, correlation, and AI/ML model.
- Short technical write-up: approach, model choice, and explain ability method.
- Dashboard/visualization showing flagged entities and evidence for each flag.

Organisation: National Technical Research Organisation

Category: Software

Theme: Cryptocurrency

Youtube Link/Video Link: Nil

Dataset Link: Nil